Math Training — M1 Cognitive Science

Session 1 — Linear Algebra I

Vectors, span, dot product, change of basis, matrices

Advanced track

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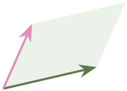
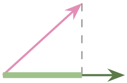
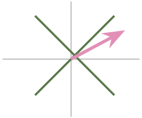
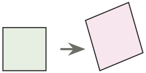
Roadmap of the session

1. Vector, linear combination, **span**
2. Dot product $\longrightarrow$ **projection**
3. **Change of basis**: same point, new axes
4. **Matrix** $=$ transformation of space

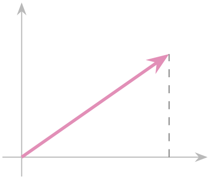

span


projection

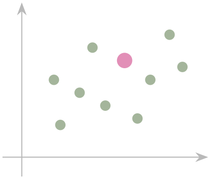

new axes


transformation

One object, three readings



arrow: length, angle



point: one row of a table

$$\mathbf{p} = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}$$

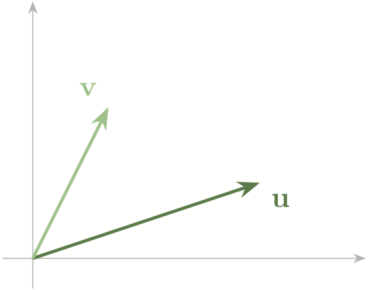
one number per measure

list: how data arrives

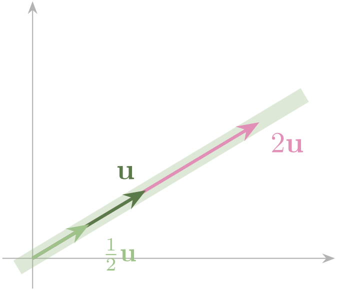
Definition

$\mathbf{v} \in \mathbb{R}^n$: an ordered list of n numbers. Two operations only — **add** slot by slot, **scale** $\lambda \mathbf{v} = (\lambda v_1, \ldots, \lambda v_n)$.

Adding and scaling, drawn



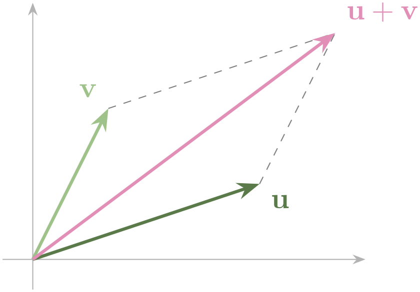
add: tip to tail



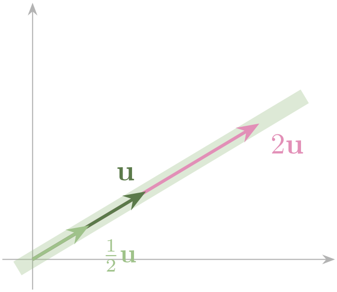
scale: never leaves the line

Left: the sum is the far corner of the parallelogram. Right: scaling slides the arrow along its own line — which is why one vector spans only a line.

Adding and scaling, drawn



add: tip to tail



scale: never leaves the line

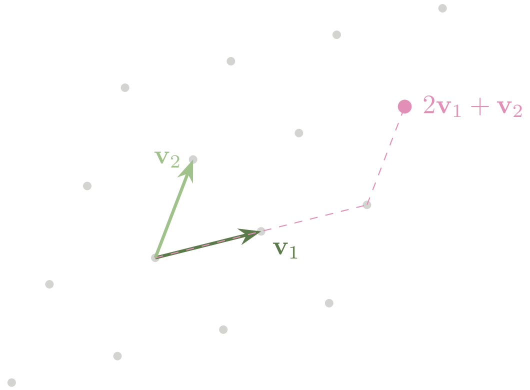
Left: the sum is the far corner of the parallelogram. Right: scaling slides the arrow along its own line — which is why one vector spans only a line.

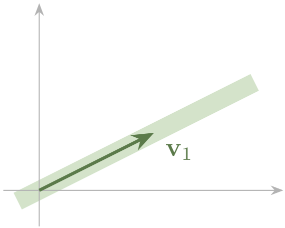
Scale each one, then add them up

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_k \mathbf{v}_k.$$

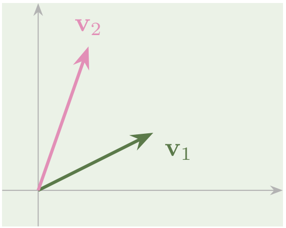
All of it is this one operation:

- span, dependence
- basis, coordinates
- matrix–vector product
- rank, eigenvectors
- principal components





one vector $\Rightarrow$ a **line**

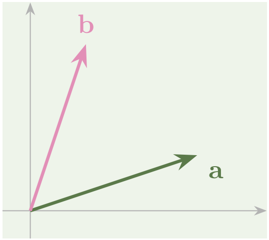


two independent $\Rightarrow$ a **plane**

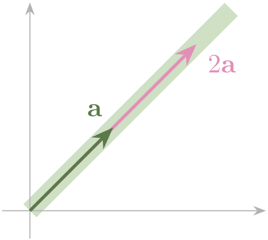
Definition

$\mathrm{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ is the set of *all* their linear combinations: a line, a plane, or a flat of higher dimension — always through the origin.

Independent, or not



independent: the whole plane



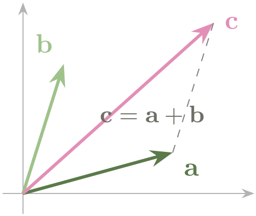
dependent: one line only

Definition

Dependent $\iff$ $\lambda_1 \mathbf{v}_1 + \cdots + \lambda_k \mathbf{v}_k = \mathbf{0}$ for some λ_i *not all zero*. Independent otherwise.

- **Useful fact**

  In $\mathbb{R}^n$: any $n+1$ vectors are dependent, and any n independent vectors span all of $\mathbb{R}^n$.



Three vectors in $\mathbb{R}^2$? Dependent.
No computation needed.

This is the fact *rank* will rest on next session.

A trap worth naming

Watch out

"They look different, so they are independent."

$$(1,2)\ \&\ (2,4) \qquad (3,3)\ \&\ (1,1)$$

Different-looking lists of numbers — each a multiple of the other.

The cure: ask for the vanishing combination explicitly. $\quad 2(1,2) - 1(2,4) = \mathbf{0}$

The running example: a data table

	RT	acc.	anx.
P_1	4	9	1
P_2	6	7	3
P_3	5	8	2
P_4	9	4	6
P_5	7	6	4

$\longleftarrow$ a **row** = one participant

$\uparrow$ a **column** = one measure, across everybody

- **Link to cognitive science**

  $P_2 = (6, 7, 3)$ is a point in $\mathbb{R}^3$; the accuracy column $(9, 7, 8, 4, 6)$ is a point in $\mathbb{R}^5$. **Same table, two different vector spaces** — and which one you are in is decided by the question you are asking.

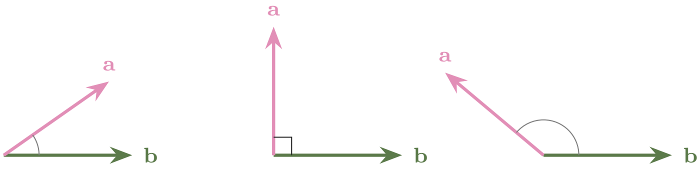
Practice — try it

1. $\mathbf{a} = (1, 2)$, $\mathbf{b} = (3, -1)$: compute $\mathbf{a} + \mathbf{b}$, $2\mathbf{a}$, $\mathbf{a} - \mathbf{b}$.
2. Independent? (i) $(1, 1)$ & $(2, 2)$; (ii) $(1, 0)$ & $(0, 1)$; (iii) $(2, -4)$ & $(-1, 2)$.
   Give a vanishing combination for each dependent pair.
3. Describe $\text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$. Is $(1, 2, 1)$ in it?

The dot product: one number, two formulas

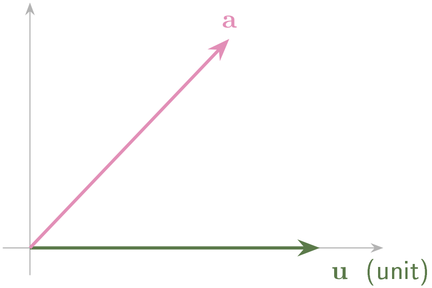
Definition

$$\mathbf{a}\cdot\mathbf{b} = \sum_{i=1}^{n} a_i b_i \;=\; \|\mathbf{a}\|\,\|\mathbf{b}\|\cos\alpha, \qquad \|\mathbf{a}\| = \sqrt{\mathbf{a}\cdot\mathbf{a}}.$$



$\mathbf{a}\cdot\mathbf{b} > 0$ $\qquad$ $\mathbf{a}\cdot\mathbf{b} = 0$ $\qquad$ $\mathbf{a}\cdot\mathbf{b} < 0$

The half usually missing: projection

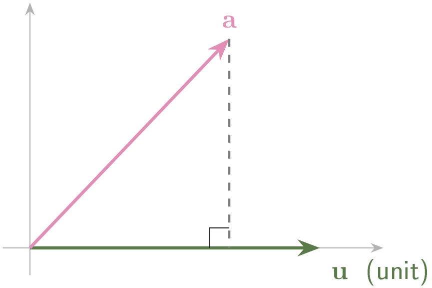


- **Useful fact**

$$\mathbf{a}\cdot\mathbf{b} = \underbrace{\left(\|\mathbf{a}\|\cos\alpha\right)}_{\text{length of the shadow}} \times \|\mathbf{b}\|, \qquad \text{and for } \textit{unit } \mathbf{u}: \quad \mathbf{a}\cdot\mathbf{u} = \text{signed shadow length.}$$

A coordinate in an orthonormal basis *is* a dot product.

The half usually missing: projection



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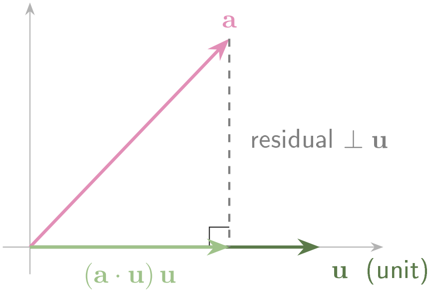


- **Useful fact**

$$\mathbf{a} \cdot \mathbf{b} = \underbrace{\left(\|\mathbf{a}\| \cos \alpha\right)}_{\text{length of the shadow}} \times \|\mathbf{b}\|, \qquad \text{and for } \textit{unit } \mathbf{u}: \quad \mathbf{a} \cdot \mathbf{u} = \text{signed shadow length.}$$

A coordinate in an orthonormal basis *is* a dot product.

Two more facts, and one warning

- **Cauchy—Schwarz**

  $|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \, \|\mathbf{b}\|$, with equality exactly when $\mathbf{a}$ and $\mathbf{b}$ are dependent.

  Divide through: $\cos \alpha \in [-1, 1]$ — the *cosine similarity*.

- **Watch out**

  The dot product is **not** bounded by 1: $(10, 0) \cdot (10, 0) = 100$.

  The bounded quantity is $\dfrac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}$.

Dot product — worked example

Example

A composite score weights the three measures: $\mathbf{w} = (1,\ 2,\ -1)$ (reward speed a little, accuracy a lot, penalise anxiety).

Two participants

$$\mathbf{p} = (1, 1, 0): \quad \mathbf{w} \cdot \mathbf{p} = 1 + 2 - 0 = 3$$

$$\mathbf{q} = (0, 1, 2): \quad \mathbf{w} \cdot \mathbf{q} = 0 + 2 - 2 = 0$$

$\mathbf{q}$ scores zero — her accuracy is exactly cancelled by her anxiety.

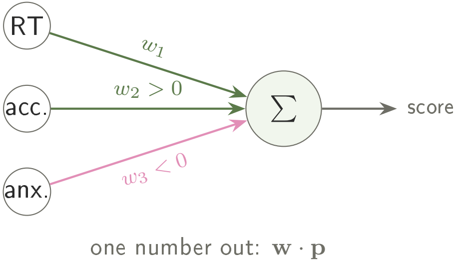
Normalised

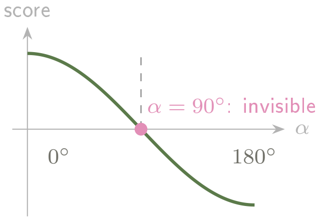
$$\|\mathbf{w}\| = \sqrt{1 + 4 + 1} = \sqrt{6} \approx 2.45$$

$$\mathbf{p} \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|} = \frac{3}{\sqrt{6}} \approx 1.22$$

$\Rightarrow$ $\mathbf{p}$'s profile *points the same way* as the scoring rule.

How a scoring rule reads a profile





Link to cognitive science

Score $= w_1 p_1 + \cdots + w_n p_n = \mathbf{w} \cdot \mathbf{p}$. Among profiles of the same size, the score follows $\cos \alpha$: maximal along $\mathbf{w}$, **exactly zero for any profile orthogonal to $\mathbf{w}$**. Such a participant is invisible to this rule — not weakly seen, invisible.

Practice — the dot product

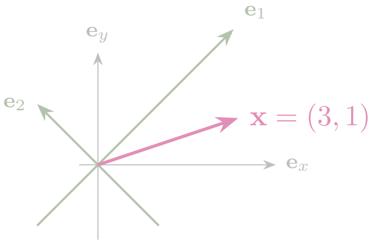
Practice — try it

1. Compute $(1, 2, 3) \cdot (4, 0, -1)$ and $\|(3, 4)\|$.
2. Are $(1, 2)$ and $(2, -1)$ orthogonal? What does that mean for a scoring rule with weights $(1, 2)$ applied to the profile $(2, -1)$?
3. Project $\mathbf{a} = (3, 1)$ onto $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$: signed shadow length, then the projection vector.

Axes you choose, not axes you inherit

Definition

A **basis**: reference axes in which every vector is written in exactly one way.
Orthonormal: the axes are perpendicular and of length one — and then every coordinate is a projection, $x_i = \mathbf{x} \cdot \mathbf{e}_i$.



The pink arrow never moves. Grey axes: $(3, 1)$. Tilted axes: $(2\sqrt{2}, -\sqrt{2})$.

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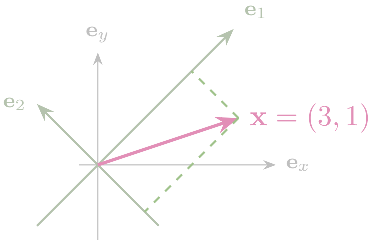

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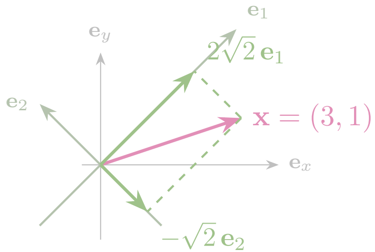


The pink arrow never moves. Grey axes: $(3, 1)$. Tilted axes: $(2\sqrt{2}, -\sqrt{2})$.

Out and back — run both on every example

$$\begin{tikzcd} \mathbf{x}\quad \text{(old axes)} \arrow[r, bend left, "\mathbf{x}_{\text{new}} = \mathbf{Q}^\top \mathbf{x}\quad \text{(dot products)}"] & \mathbf{x}_{\text{new}}\quad \text{(new axes)} \arrow[l, bend left, "\mathbf{x} = \mathbf{Q}\,\mathbf{x}_{\text{new}}\quad \text{(linear combination)}"] \end{tikzcd}$$

- **Useful fact**

  $\mathbf{Q} = (\mathbf{e}_1\ \mathbf{e}_2)$ holds the new axes as *columns*. Orthonormality is exactly $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}$, i.e. $\mathbf{Q}^{-1} = \mathbf{Q}^\top$: no inversion is ever needed.

Two checks that catch everything

- **Useful fact**

  1. **Rebuild.** Reconstruct the original vector from its new coordinates.
  2. **Compare lengths.** $x_1^2 + x_2^2 = \|\mathbf{x}\|^2$.

The second is Pythagoras — and it is exactly what fails for non-perpendicular axes.

Example

Axes rotated by $45°$: $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-1,1)$. Express $\mathbf{x} = (3,1)$.

$$x_1 = \mathbf{x}\cdot\mathbf{e}_1 = \frac{1}{\sqrt{2}}(3+1) = 2\sqrt{2}, \qquad x_2 = \mathbf{x}\cdot\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-3+1) = -\sqrt{2}.$$

Rebuild

$$2\sqrt{2}\,\mathbf{e}_1 - \sqrt{2}\,\mathbf{e}_2 = (2,2)+(1,-1) = (3,1) \checkmark$$

Lengths

$$x_1^2 + x_2^2 = 8 + 2 = 10 = \|\mathbf{x}\|^2 \checkmark$$

The two wrong answers

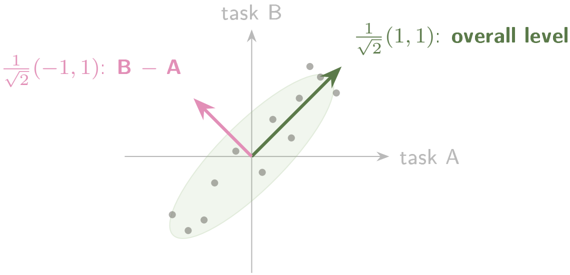
- **Watch out**

$$(3\sqrt{2},\ -2\sqrt{2}) \qquad \text{and} \qquad \sqrt{2}\,(\mathbf{e}_1 - 2\mathbf{e}_2)$$

First: the projections were never taken.
Second: the two coefficients were swapped.

Both are caught instantly by **rebuilding** and by **comparing lengths**.

Why choosing axes is half the job



Link to cognitive science

Each dot is a participant's score on two similar tasks (centred). "Task A" and "task B" are the axes the *experiment* handed us; "how good overall" and "which task is easier for them" are the axes the *question* wants. In them the covariance $\left(\begin{smallmatrix} v & c \\ c & v \end{smallmatrix}\right)$ becomes diagonal — the two new scores are uncorrelated.

Practice — try it

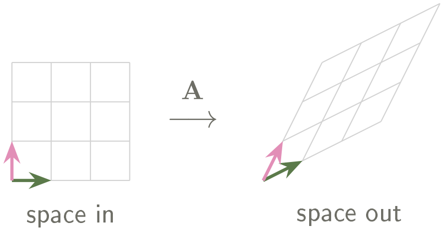
1. Show that $\{\frac{1}{\sqrt{2}}(1,1),\ \frac{1}{\sqrt{2}}(1,-1)\}$ is orthonormal.
2. Write $\mathbf{x} = (2,0)$ in the basis $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-1,1)$, then rebuild it as a check.
3. Give an orthonormal basis whose first axis points along $(3,4)$.

A matrix is a machine that moves vectors

Definition

$\mathbf{A}$ of size $m \times n$ sends $\mathbf{x} \in \mathbb{R}^n$ to $\mathbf{A}\mathbf{x} \in \mathbb{R}^m$; output slot i is (row i) $\cdot$ $\mathbf{x}$:

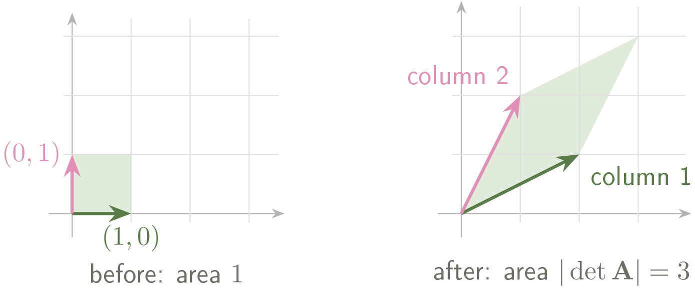
$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{pmatrix}.$$



Rotate, stretch, shear or squash — origin fixed, straight lines straight, grid lines evenly spaced.

A matrix is where the axes land

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{A}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \mathbf{A}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$



The columns are where the axis arrows land; everything else follows. This same $\mathbf{A}$ returns in Session 2.

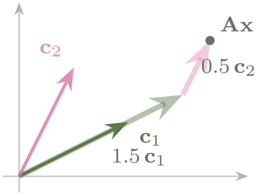
- **Intuition**

$$\mathbf{A}\mathbf{x} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2$$

The output is a linear combination of the *columns*.

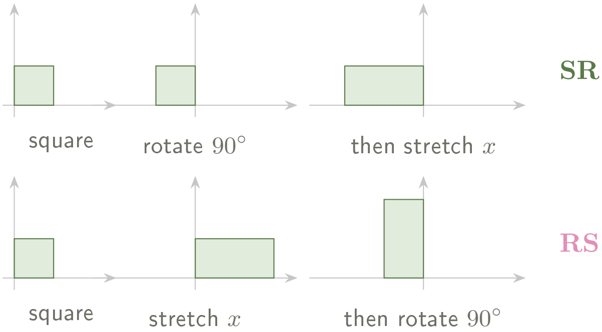
$\Rightarrow$ the set of possible outputs is exactly span(columns).

That is what makes *rank* obvious next session.



$\mathbf{x} = (1.5, 0.5)$

Order matters: $\mathbf{AB} \neq \mathbf{BA}$



Same two operations, opposite order, different shape. In $\mathbf{AB}$ the *right-hand* matrix acts first; $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$ reverses too.

Example

Recode two task scores as *mean* and *difference*: $\mathbf{M} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -1 & 1 \end{pmatrix}$, scores $\mathbf{s} = (8, 4)$.

$$\begin{array}{c} \boxed{\begin{array}{l} \text{task A} = 8 \\ \text{task B} = 4 \end{array}} \\ \Big\downarrow \mathbf{M} \\ \boxed{\begin{array}{c} \text{mean} = 6 \\ \text{B} - \text{A} = -4 \end{array}} \end{array}$$

By rows

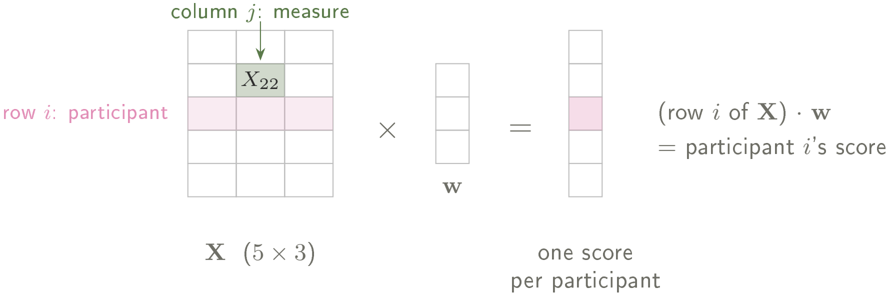
$$\mathbf{M}\mathbf{s} = \begin{pmatrix} (\frac{1}{2}, \frac{1}{2}) \cdot (8, 4) \\ (-1, 1) \cdot (8, 4) \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$$

By columns

$$8\begin{pmatrix} \frac{1}{2} \\ -1 \end{pmatrix} + 4\begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix} \checkmark$$

Same participant, two numbers in, two numbers out — and the second pair is the one you would actually analyse.

The data matrix



- **Link to cognitive science**

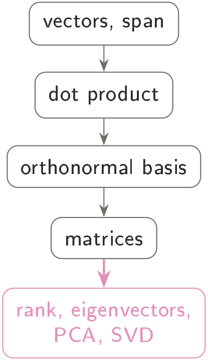
  X_{ij}: **participant** i, **measure** j — rows are people, columns are variables. Then $\mathbf{X}\mathbf{w}$ applies one scoring rule to *everybody at once*, in a single product. Predicting an outcome from several measures is exactly this: the linear part of a regression is $\mathbf{X}\mathbf{w}$.

Practice — try it

1. Compute $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$; say in words what this matrix does.
2. Compute $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 5 \\ -2 \end{pmatrix}$. Which geometric operation is it?
3. Compute $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ and the same product the other way. Comment.

Session 1 — what we built

- **Span** = what a set of vectors can build; **independence** = no redundancy.
- **Dot product** ⇒ length, angle, and the **projection** picture.
- **Change of basis**: out with $\mathbf{Q}^\top$, back with $\mathbf{Q}$ — checked twice.
- **Matrix** = transformation: rows to compute, columns to understand.



Next session: what a transformation does to space — and how many dimensions a data table really has.